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**B.Tech. Degree IV Semester Special Supplementary Examination
February 2020**

**ME 15-1405 HYDRAULIC MACHINERY
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Describe Rayleigh's method of dimensional analysis.
- (b) Define: (i) Model (ii) Prototype (iii) Model analysis (iv) Hydraulic similitude.
- (c) Show that the efficiency of free jet striking normally on a series of flat plates mounted on the periphery of a wheel can never exceed 50%.
- (d) Differentiate between impulse and reaction turbines.
- (e) Define specific speed of a turbine. Derive an expression for it. Explain its significance.
- (f) What is cavitation? State its effects on the performance of hydraulic machines. How will you prevent cavitation?
- (g) What is an air vessel? Describe the functions of air vessel for reciprocating pumps.
- (h) Draw a neat sketch and explain the working of hydraulic press.
- (i) Explain the working of hydraulic accumulator. Obtain an expression for the capacity.
- (j) In a hydraulic coupling, the speeds of the driving and driven shafts are 800 rpm and 780 rpm respectively. Find (i) the efficiency of the hydraulic coupling (ii) the slip of the coupling.

**PART B**

(4 × 10 = 40)

- II. (a) State and explain Buckingham's π theorem. (3)
- (b) The pressure difference Δp in a pipe of diameter D and length l depends on the velocity V , viscosity μ , density ρ and roughness k . Using Buckingham's π theorem, obtain an expression for Δp . (7)

OR

- III. A jet of water having a velocity of 30 m/s strikes a series of vanes mounted on a wheel which is rotating at 200 rpm. The jet makes an angle of 20° with the tangent to the wheel at inlet and leaves the wheel with a velocity of 5 m/s at an angle 130° to the tangent to the wheel at the outlet. Water is flowing from outward in radial direction. The outer and inner radii of the wheel are 0.5 m and 0.25 m respectively. Find (i) Vane angle at inlet and outlet (ii) Work done per unit weight of water (iii) Efficiency of the wheel. (10)

- IV. Design a Pelton wheel for the following specifications: (10)
 Shaft Power = 735.75 kW
 Head = 200 m
 Speed = 800 rpm
 Overall efficiency = 86%
 The jet diameter should not exceed one-tenth of the wheel diameter.

OR

(P.T.O.)

- V. (a) What is draft tube? Explain the functions of draft tube. (3)
 (b) A conical draft tube having inlet and outlet diameters 1 m and 1.5 m, discharges water at outlet with a velocity of 2.5 m/s. The total length of the draft tube is 6 m and 1.2 m of the length of the draft tube is immersed in water. If the atmospheric pressure head is 10.3 m of water and loss of head due to friction in the draft tube is equal to $0.2 \times$ velocity of head at the outlet of the tube, find (i) Pressure head at inlet (ii) efficiency of the draft tube. (7)
- VI. A three stage centrifugal pump has impeller 40 cm in diameter and 2.5 cm wide at outlet. The vanes are curved back at the outlet at 30° and reduce the circumferential area by 15%. The manometric efficiency is 85% and the overall efficiency is 75%. Determine the head generated by pump when running at 1200 rpm and the discharge is $0.06 \text{ m}^3/\text{s}$. Also find the shaft power. (10)
- OR**
- VII. Explain the effect of acceleration and friction in suction and delivery pipes on indicator diagrams of a reciprocating pump. (10)
- VIII. Explain the principle and working of the following with the help of neat sketches: (10)
 (i) Hydraulic crane
 (ii) Hydraulic torque converter
 (iii) Hydraulic coupling.
- OR**
- IX. A hydraulic lift is required to lift a load of 12 kN through a height of 10 m, once in every 1.75 minutes. The speed of the lift is 0.75 m/s. During work stroke of the lift, water from the accumulator and the pump at a pressure of 400 N/cm^2 is supplied to the lift. If the efficiency of the pump is 80% and that of the lift is 75%, find the power required to drive the pump and minimum capacity of the accumulator. Neglect frictional losses in the pipe. (10)
